Tutorial Session 2

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Problem 6 [7 pts]. Label the following statements as true or false.

(a) If V is a vector space and W is a subset of V that is a vector space, then W is a subspace of V.

$V = \mathbb{R}^2$ $W = \{(x, 1)\}$ $(x_1, 1) + (x_2, 1) = (x_1 + x_2, 1)$

$c \cdot (x, 1) = (cx, 1)$

Problem 20 [16 pts]. Consider a system of linear equations

$$\begin{aligned}
(a_1+b)x_1 + a_2x_2 + a_3x_3 + \cdots + a_nx_n &= 0\\
a_1x_1 + (a_2+b)x_2 + a_3x_3 + \cdots + a_nx_n &= 0\\
a_1x_1 + a_2x_2 + (a_3+b)x_3 + \cdots + a_nx_n &= 0\\
&\cdots\\
a_1x_1 + a_2x_2 + a_3x_3 + \cdots + (a_n+b)x_n &= 0
\end{aligned}$$

where $\sum_{i=1}^{n} a_i \neq 0$.

Determine the relationship between $a_1, a_2, \cdots, a_n$ and b such that

(a) This linear system has only a zero solution. (8 pts)

(b) This system has nonzero solutions. In this case, determine a solution set. (8 pts)

Problem 1 Prove that the following two axioms:

- $\exists \vec{0} \in V$ such that $\vec{\alpha} + \vec{0} = \vec{0} + \vec{\alpha} = \vec{\alpha}$.
- $\forall \vec{\alpha} \in V$, $\exists \vec{\beta} \in V$ such that $\vec{\alpha} + \vec{\beta} = \vec{\beta} + \vec{\alpha} = \vec{0}$.

Is equivalent with the following statement:

- $\forall \vec{\alpha}, \vec{\beta} \in V$, $\exists \vec{x} \in V$ such that $\vec{\alpha} + \vec{x} = \vec{\beta}$.

When trying to prove the equivalence, you may use the rest of the six axioms of vector space. Strictly speaking, we are proving the equivalence conditioned on the other six axioms.

$\Rightarrow \vec{x} = (-\vec{\alpha}) + \vec{\beta}$

$\Leftarrow \exists \vec{0}_\alpha \quad \vec{\alpha} + \vec{0}_\alpha = \vec{\alpha}$

$\forall \vec{\beta}, \vec{\beta} + \vec{0}_\alpha = \vec{\beta}$

$\exists \vec{\gamma} \quad \vec{\beta} = \vec{\alpha} + \vec{\gamma} = \vec{\gamma} + \vec{\alpha}$

$\therefore \vec{\beta} + \vec{0}_\alpha = \vec{\gamma} + \vec{\alpha} + \vec{\alpha} = \vec{\gamma} + \vec{\alpha} = \vec{\beta}$

$=$

$\forall \vec{\alpha}, \exists \vec{\beta}, \vec{\alpha} + \vec{\beta} = \vec{0}$

Problem 2 Consider the following subset V of $\mathbb{R}^{3\times 3}$: The elements of V are of the form

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix},$$

such that the sum of the elements in each row, column, the main diagonal, and the anti-diagonal are equal. For example, the following matrix is an element of V:

$$\begin{pmatrix} 6 & 1 & 8 \\ 7 & 5 & 3 \\ 2 & 9 & 4 \end{pmatrix}.$$

1. Prove that V is a linear subspace of $\mathbb{R}^{3\times 3}$.
2. Determine the dimension and a basis of V.

Hint: Can you find a linear transformation such that V is the null space?

$W_1, W_2, \ldots, W_k$

Problem 3 Let V be a vector space of finite dimension n, and W_1, W_2 be two subspaces of V with dimension no larger than $m < n$. Prove that there exists a linear subspace W of V that is of dimension $n - m$, and $W \cap W_1 = W \cap W_2 = \{\vec{0}\}$.

Hint: Recall from homework we proved that $W_1 \cup W_2$ is a subspace of V if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$. What does this tell you for the case $m = n - 1$?

$m \times m$

$n \times n$

$W_1 \cup W_2 \subsetneq V$

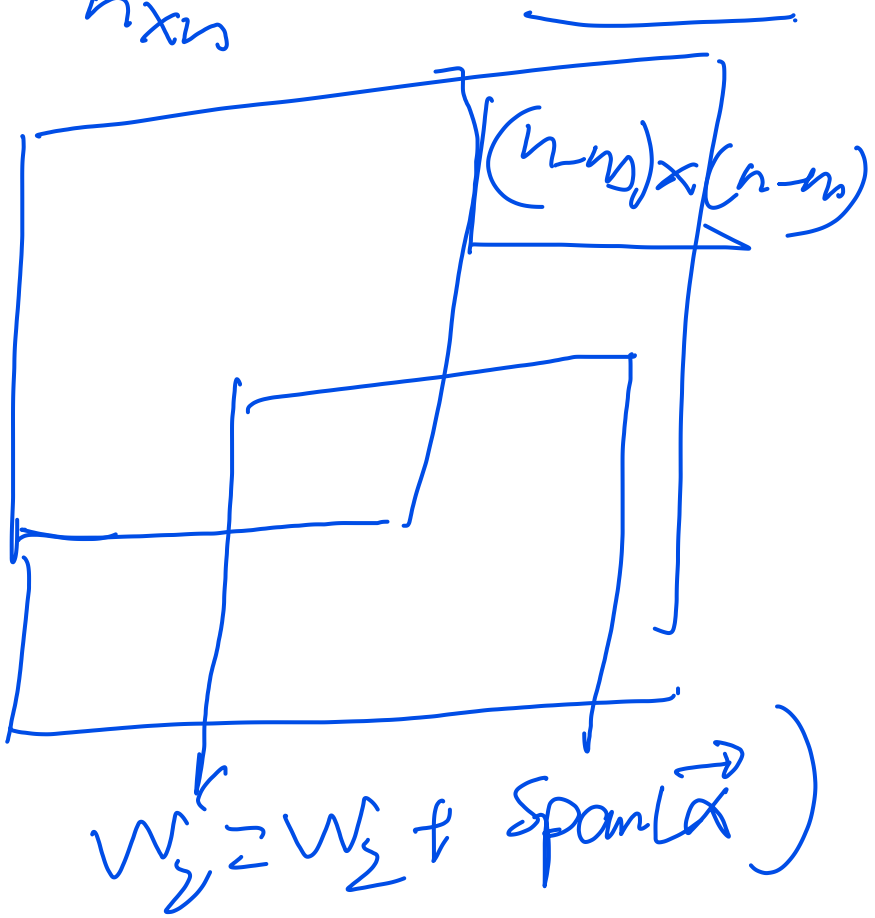
$\exists\, \vec{\alpha},\ \vec{\alpha} \in V / (W_1 \cup W_2)$

$\mathrm{span}(\vec{\alpha}) = W$ c.$\vec{\alpha}$

$W \cap W_1 = W \cap W_2 = \{\vec{0}\}$

$n - m$

$W_1' = W_1 + \mathrm{span}(\vec{\alpha})$



Problem 4 Given vector space V. We call a linear transformation $T : V \to V$ idempotent if $T^2 = T$. Prove that if $\sigma, \tau, \sigma + \tau : V \to V$ are idempotent linear transformations from V to V, then $\sigma\tau = 0$ is the zero transformation.

finite-dimensional

Problem 5 Assume V_1, V_2, V_3 are three vector spaces defined on the same field $\mathbb{F}$. Let $\varphi : V_1 \to V_2, \psi : V_1 \to V_3$ be two linear transformations. Prove that the following statements are equivalent:

- There exists linear transformation $\sigma : V_2 \to V_3$ such that $\psi = \sigma \circ \varphi$.
- The null space of φ is a subspace of the null space of ψ.

A further question: Can you identify a necessary and sufficient condition about φ, ψ and the vector spaces V_1, V_2, V_3 such that not only σ exists, but also σ is unique?

